

How the CDR model works

For every $\sim 1 \text{ km}^2$ pixel the model asks: *how much CO_2 does one tonne of crushed basalt durably remove here?* A chemical ceiling is reduced by fractional “haircuts,” scaled by the application rate, then netted against the project’s own emissions.

1. Chemical ceiling (stoichiometry). CaO/MgO each consume 2 mol CO_2 :

$$\text{CDR}_{\text{max}} = \left(\underbrace{0.10}_{\text{CaO}} \cdot \frac{2 \cdot 44.01}{56.08} + \underbrace{0.07}_{\text{MgO}} \cdot \frac{2 \cdot 44.01}{40.30} \right) \cdot 1000 \approx 310 \text{ kg CO}_2/\text{t basalt}.$$

2–4. Fractional haircuts (reactivity, grain size, climate, soil pH, carbonate re-precipitation):

$$\text{CDR}_{\text{per t}}(px) = \text{CDR}_{\text{max}} \times \underbrace{0.20}_{\text{reactive}} \times \underbrace{\left(\frac{100}{d}\right)^{0.5}}_{\text{grain}} \times \underbrace{f_{\text{MAT}} f_{\text{MAP}} f_{\text{pH}}}_{\text{climate, capped}} \times \underbrace{(1 - p_{\text{ped}})}_{\text{exported}}.$$

$$f_{\text{MAT}} = \text{clamp}\left(\frac{\text{MAT}}{11^\circ\text{C}}, 0.3, 3\right), \quad f_{\text{MAP}} = \text{clamp}\left(\frac{\text{MAP}}{1000 \text{ mm}}, 0.3, 3\right), \quad f_{\text{pH}} : \text{triangular, peak} \approx \text{pH } 5.$$

5–7. Yield, net of emissions, revenue.

$$\text{CDR}_{\text{gross}} = \text{rate} \cdot \frac{\text{CDR}_{\text{per t}}}{1000}, \quad \text{CDR}_{\text{net}} = \max\left(0, \text{CDR}_{\text{gross}} - \text{rate} \cdot \frac{\text{LCA}_{\text{per t}}}{1000}\right),$$

$$\text{revenue} = \text{CDR}_{\text{net}} \cdot (p_{\text{carbon}} - \text{MRV}), \quad \text{NPV regime: } \times 5\text{-yr phasing factor} \approx 0.79.$$

In short: chemistry sets the ceiling ($\sim 310 \text{ kg/t}$); geography (temperature, rainfall, acidity, aridity) decides how much of it you actually capture ($\sim 80 \text{ kg/t}$ at the reference climate).

Key references for methodology & coefficients

Chemistry & stoichiometry

Lewis et al. 2021, *Appl. Geochem.* 132:105023

Beerling et al. 2020, *Nature* 583:242 (SI)

Reactive fraction (0.20)

Lewis et al. 2021 — load-bearing calibration

Grain size & grinding ($\beta=0.5$, $\alpha=1.2$)

Strefler et al. 2018, *ERL* 13:034010

Climate scaling

Kanzaki et al. 2023, *PNAS Nexus* 2:pgad059

Baek et al. 2023, *Earth's Future* 11

pH dependence

White & Brantley 2003, *Chem. Geol.* 202:479

Brantley et al. 2008, *Kinetics of Water-Rock Interaction*

Pedogenic-carbonate deduction

Renforth 2019, *Nat. Commun.* 10:1401

Reershemius & Suhrhoff 2024, *GCB* 30:e17025

UNEP 1992 (aridity bands); Zomer et al. 2022, *Sci. Data* 9:409

Net CDR / lifecycle emissions

Eufrazio et al. 2022, *Commun. Earth Environ.* 3:106

EEA EMEP 2023; UK BEIS 2023; Nemecek & Schnetzer 2011

Time phasing & MRV

Kanzaki et al. 2023; Levy et al. 2024, *ES&T* 58:17215

Application rate & targeting

Beerling 2020; Baek 2023; Kantola et al. 2023, *GCB* 29:7012

Aramburu Merlos et al. 2023, *Geoderma* 432:116421

Full catalogue (S1–S37) with DOIs: [docs/erw-parameters.md](#)